SCALE Grant Proposal: Private Lending for Aleo

PWN DAO Foundation · 2 November 2026
Request: $85,000 · Delivery: 17 weeks from project start

1. Project Overview

SCALE is peer-to-peer lending infrastructure for Aleo. Asset holders can borrow without selling; lenders choose the rate, duration and collateral requirement. Both sides can post public or private offers and accept multiple fills. Each fill creates a separate loan. Loans use no price oracles or price-triggered liquidations.

The grant takes the proof of concept to an audited mainnet release, starting with ALEO collateral and USDCx credit. Deliverables are Leo programs, a TypeScript SDK, a reference app and integration documentation, released under GPL-3.0 with zero protocol fees in V1. Wallets and applications can embed the SDK; developers can extend the programs or deploy their own markets.

2. Team Background

PWN DAO Foundation has built lending infrastructure for over five years. PWN is deployed on 13 EVM mainnets and has processed over $10m in loan volume for more than 5,000 users. The team also builds OWN, crypto-backed mortgage infrastructure using PWN's lending primitives.

- **Josef Jelacic** — PWN DAO and OWN founder; previously at the Ethereum Foundation. Architecture and design oversight. LinkedIn.
- **Šimon Kozák** — founding engineer and CTO at PWN DAO and OWN. Protocol and application development. LinkedIn · GitHub.
- **Jan Mikšík** — engineer with four years building DeFi lending applications. Protocol and application development. LinkedIn · GitHub.

Šimon and Jan will jointly deliver the programs, SDK and reference app.

3. Market Potential & Demand

The initial audience is ALEO holders seeking liquidity without selling, stablecoin holders willing to lend, and funds or treasuries arranging bilateral credit. USDCx and USAD had combined on-chain supply of approximately **$4.71m at peg on 2 November 2026**. USDCx state · USAD state.

For broader market context, Galaxy Research reported $56.16bn in crypto-collateralized loans at Q2 2026, including $20.43bn in DeFi. Federal Reserve researchers estimated $138bn in US securities-based loans in Q1 2024. These figures describe available assets and adjacent market scale, not demonstrated demand for SCALE or amounts that can be added into one TAM. Pilot lenders, borrowers and completed loans will validate demand.

Wallets, payment applications and treasury tools can integrate lending through the SDK. Shield wallet, USAD, Pondo/pALEO and Beta Staking/stALEO are integration targets, not confirmed partnerships. Liquid-staking collateral follows the initial production pair. Lenders retain default and collateral-price risk.

4. System Design & Specification

Offers and settlement

The offer maker escrows their side: credit for a lender offer, collateral for a borrower offer. The accepting party supplies the other asset. Each offer sets total capacity, minimum and maximum fill amounts, fixed pricing, duration and expiry. Each fill creates an independent loan, starting its own maturity clock and assigning its own lender and collateral. Public borrower terms support discovery; acceptance also needs an off-chain opening with the borrower's payout details.

<table>
<tr><td>Lender posts
Credit escrowed</td><td>Borrower posts
Collateral escrowed</td></tr>
<tr><td colspan="2">Public listing or private invitation · Repeated fills within remaining capacity</td></tr>
<tr><td>Borrowers accept one or more fills</td><td>Lenders accept one or more fills</td></tr>
<tr><td colspan="2">Each fill creates a separate loan atomically
Collateral allocated · Principal paid to borrower</td></tr>
<tr><td>Repayment settles first
Lender receives repayment
Borrower recovers that loan's collateral</td><td>Claim settles first, at/after maturity
Lender receives that loan's collateral</td></tr>
</table>

After maturity, the lender may claim collateral. Repayment remains possible until that claim settles; the first successful settlement closes the loan. Transfers and accounting succeed together or revert. Repayment and collateral recovery do not replenish offer capacity. Expiry blocks new fills; cancellation refunds unused escrow without changing existing loans. Per-loan collateral rounds upward; fills must fit both remaining principal and available collateral, and unused remainders are refundable.

Components and assets

- **Hub/Config:** authorize offer modules and stop new originations. Production controls cannot redirect or block settlement of existing loans.
- **Offer modules:** validate terms and acceptance rights, manage lender credit and withdrawals, and keep borrower-offer repayments separate for each matched lender.
- **Loan core:** holds unused borrower collateral reservations and active collateral, creates loans and enforces repayment, default claims and one-time settlement.

- **Token programs:** transfer assets through program-controlled public balances; user payouts are private records. Programs hold no spendable private records.
- **SDK/app:** prepare transactions, discover offers, verify commitments, show remaining capacity and individual loans, and manage encrypted payloads and recovery.

V1 has no administrative asset-admission process. Assets must match supported interfaces; settlement binds the exact token program and accounting is isolated per token. Module authorization does not approve assets. The PoC covers native ALEO, ARC-20 and the USDCx/USAD compliance profiles. NFTs, rebasing and fee-on-transfer tokens are outside V1. Issuer freeze, burn, pause and upgrade powers may disrupt settlement; M1 defines the production deficit policy.

Privacy

Public offers disclose advertised terms. Private offers share an encrypted payload with a designated counterparty and commit to it on-chain. Acceptance proves the opening and counterparty authorization; that counterparty may fill repeatedly. Payload encryption and recovery are SDK deliverables.

Data	Visibility and limits
Wallet identity	Private inputs, records and salted authorization avoid direct public wallet-to-loan fields. Counterparties receive necessary payout details. Correlation and counterparty disclosure remain possible.
Private offers	Complete payloads are not publicly listed. Commitments, status, expiry, necessary capacity data and escrow movements remain visible; assets and amounts may be revealed before acceptance.
Active loans	Submitted loan amounts, rate and duration are public even if acceptance is rejected. Created loans also expose maturity, status and program balances. Confidential settlement and hidden amounts are outside V1.
Compliance and proving	USDCx/USAD compliance observers receive designated transfer receipts. Delegated provers may see inputs unless a verified confidential setup protects them.

The SDK requires private fees without silent public-fee fallback, fresh ownership/opening salts, encrypted backups, and disclosure of one loan without an account-wide view key. The privacy review covers successful and rejected transactions, indexers, proving services and timing/amount correlation. Records and view keys · Stablecoin visibility.

Funding experiment and implementation evidence

M1 will validate optional funding at loan creation using supported public token allowances. This exposes the lender's public funding address and transfer amount, leaves balances unreserved, and requires handling revocation or insufficient funds. Native ALEO borrower collateral still needs

borrower authorization at final execution. The deliverable is a published feasibility result; funded offers remain the baseline.

The public-testnet predecessor executed the lender-funded lifecycle with real proofs. The published multi-token PoC has local devnode lifecycle, accounting and adversarial tests. Production work includes borrower reservations and private-offer commitments. Local proof generation is distinct from node verification. Proof-enabled public-network validation, live USDCx operation, wallet integration, encrypted sharing and recovery remain grant work. The production core will be audited and immutable, replacing PoC timelocked upgrades. Installments, refinancing and crowdfunded pooled loans are outside V1.

5. Funding & Milestones

Milestone	Completion	Amount
M1 · Core and interfaces	Week 7	$35,000
M2 · SDK and discovery	Week 10	$15,000
M3 · Reference app	Week 13	$15,000
M4 · Audit and launch	Week 17	$20,000
Total	**17 weeks**	**$85,000**

M1 · Core and interfaces

Deliver production programs, both offer directions and visibility modes, compatible assets without admission, live USDCx integration, threat/disclosure models and the allowance experiment. On public testnet demonstrate one lender offer serving two borrowers, one borrower offer funded by two lenders, repeated authorized private fills and unauthorized rejection. Publish proof-enabled lifecycle, property and fuzz tests covering allocation/rounding, cancellation, expiry, overfill, replay, settlement races, administration and token failures; document compliance outputs and privacy findings. Freeze reviewed interfaces for audit.

M2 · SDK and discovery

Publish the TypeScript package and integration docs for offer creation, fills, discovery, encrypted sharing/import, repayments, claims, cancellation and withdrawals. Include private fees, current compliance proofs and position recovery. Acceptance: complete the testnet lifecycle without a proprietary backend, verify imported commitments, recover from backup, and report stale proofs or failed transactions with actionable errors.

M3 · Reference app

Deliver wallet connection, public browsing, private invitations, both offer directions, fill sizing, remaining capacity, individual positions and selective disclosure. External testers complete lender/borrower journeys without team assistance and disclose one loan without an account-wide view key. Publish usage docs and fix blocking usability findings. Agree tester count, wallets and usability criteria with Aleo before M3 starts.

M4 · Audit and launch

The $20,000 milestone covers independent audit, remediation, retesting and launch; audit preparation starts after M1. Publish findings and remediation status, resolving material issues before mainnet deposits. Release immutable programs and the final SDK, with program IDs, source/build references, constructor policies and operational docs. Complete a mainnet loan with external participants through origination, repayment and settlement.

If a material external dependency blocks delivery, agree revised scope and timing with Aleo. Six-month targets: 50 originated loans, $250,000 cumulative principal measured in USD, and two third-party SDK integrations.